Name: ____________________

Class: ____________________

Date: ____________________

Mr. Rawson • GCSE Physics

Kinematics L1 — Speed and $s = vt$

L1 homework *— 10 P: set Tue 8 Sep, due Mon 14 Sep · 10 R: set Thu 10 Sep, due Fri 11 Sep.*

1 The equation

Speed is $\frac{\text{distance}}{\text{time}}$, and it is a **scalar**. The exam writes that relationship as $s = v\,t$ — s distance in m, v speed in m/s, t time in s. Rearranged it gives the other two: $v = \frac{s}{t}$ and $t = \frac{s}{v}$.

Write the equation first, then substitute, then answer with a unit — marks are given for all three.

2 Typical speeds — learn these

You are expected to recall roughly how fast ordinary things move, and to use them to check whether an answer is believable. How fast someone walks, runs or cycles depends on **age, terrain, fitness and distance**; the speed of **sound** and the **wind** vary too.

walking	~1.5 m/s	car	~25 m/s
running	~3 m/s	train	~55 m/s
cycling	~6 m/s	aeroplane	~250 m/s
sound in air ~330 m/s			

3 The race

On 22 August 2026 a humanoid robot ran 100 m in **9.39 s**. Usain Bolt's world record, set in 2009, is **9.58 s**.

1) Calculate the average speed of each, to 3 significant figures.

Robot $= 100/9.39 =$ **10.6** m/s; *Bolt* $= 100/9.58 =$ **10.4** m/s

2) Explain whether either is faster than a typical cyclist.

Both are — a cyclist is about 6 m/s and both average over 10 m/s.

4 Practice

3) Calculate the average speed of a car that travels 1500 m in 60 s. Is your answer a believable car speed?

$v = 1500/60 =$ **25** m/s *— exactly the typical car speed, so it is believable.*

4) Calculate how far a train travels in 120 s at 55 m/s.

$s = vt = 55 \times 120 =$ **6600** m *(6.6 km).*

5) Calculate how long sound takes to travel 1650 m through air.

$t = s/v = 1650/330 =$ **5** s.

6) Convert a car's speed of 90 km/h into m/s, then say whether it is faster than the typical car speed above.

90 km/h $= 90000\,\text{m} \div 3600\,\text{s} =$ **25** m/s *— the same as the typical car speed.*